US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12402598
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Palmer; Jay et al.

Animal drinking valve

Abstract

An improved animal drinking valve is disclosed. The valve may be configured to penetrate a penetrable container containing a liquid for consumption by an animal. The valve also may include an internal actuator assembly including a spring member that biases an actuator member into liquid-tight interfacing with a sealing member situated within a bore of the valve. In this arrangement, the actuator assembly may provide the valve with two-way actuation; that is, the valve may be openable to allow liquid to flow therethrough when sufficient force is applied in directions perpendicular and/or parallel to a distal end of the actuator member. Also, the valve may be of dual-material construction, having a plastic body and a metal actuator member, improving resistance to chewing by animals and other physical degradation. Furthermore, the valve optionally may include one or more exterior fin-like projections that facilitate installation of the valve in a host platform.

Inventors: **Palmer; Jay (Brunswick, ME), Dioli; Paul (Yarmouth, ME)**

Applicant: **ClearH2O, Inc.** (Portland, ME)

Family ID: **96175657**

Assignee: **ClearH2O, Inc. (Portland, ME)**

Appl. No.: **18/800462**

Filed: **August 12, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250212841 A1	Jul. 03, 2025

Related U.S. Application Data

us-provisional-application US 63615846 20231229

Publication Classification

Int. Cl.: **A01K7/06** (20060101); **A01K1/035** (20060101)

U.S. Cl.:

CPC **A01K7/06** (20130101); **A01K1/0356** (20130101);

Field of Classification Search

CPC: A01K (39/0213); A01K (1/0356); A01K (7/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3611999	12/1970	Hey	119/72.5	A01K 7/06
3831558	12/1973	Forbes	119/73	A01K 7/06
4402343	12/1982	Thompson	251/339	A01K 7/06
4790264	12/1987	Lack	119/72.5	A01K 7/06
6058881	12/1999	Thompson	119/72.5	A01K 39/0213
6986324	12/2005	Gabriel et al.	N/A	N/A
7303713	12/2006	Gabriel et al.	N/A	N/A
7546816	12/2008	Gabriel et al.	N/A	N/A
7677531	12/2009	Hollars	251/339	F16K 25/00
7937836	12/2010	Gabriel et al.	N/A	N/A
9732882	12/2016	Gabriel et al.	N/A	N/A
9814213	12/2016	Lane	N/A	A01K 7/06
10716293	12/2019	Lane	N/A	A01K 7/06

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2578149	12/1984	FR	A01K 7/06
WO-0052367	12/1999	WO	A01K 7/06

OTHER PUBLICATIONS

Hydropac, Avidity Science, 2024, https://www.avidityscience.com/water-systems/animal-watering/hydropac, 16 pages. cited by applicant
Hydropac, A Better System, Watering for Today, Hydropac Lab Products, 2006, 16 pages. cited by applicant

---

Primary Examiner: Perry; Monica L

Attorney, Agent or Firm: Hayes Soloway PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application claims the benefit of U.S. Provisional Patent Application No. 63/615,846, titled "Animal Drinking Valve," filed on Dec. 29, 2023, which is herein incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to valves and, more particularly, to animal drinking valves.

BACKGROUND

(2) Animal drinking valves are routinely used to provide a means by which an animal may hydrate itself. Dispensation of hydration through such a valve may be performed under the action of gravity or via pressurization, such as by a pump system in communication with the valve. The overall size and flowthrough capability of a given valve may be selected based on the specific animal(s) intended to use such valve. For instance, smaller valves may be employed for small animals like rodents, rabbits, and small primates, whereas larger valves may be employed for large animals like dogs, pigs, and large primates. Some animal drinking valves are constructed for use in-cage for an occupant animal, whereas others may be constructed for use in larger enclosures or in an open-air environment.

SUMMARY

(3) The subject matter of this application may involve, in some cases, interrelated products, alternative solutions to a particular problem, and/or a plurality of different uses of a single system or article.

(4) One example embodiment provides an animal drinking valve configured to penetrate a penetrable container to allow a liquid to flow from the penetrable container through the valve. The valve includes a body member. The valve further includes an assembly disposed within a bore of the body member. The assembly includes a sealing member. The assembly further includes an actuator member extending through the sealing member. The assembly further includes a spring member operatively interfaced with a first portion of the actuator member so as to bias a second portion of the actuator member toward the sealing member to effectuate a liquid-tight sealing of the valve. The actuator member is movable within the bore against the bias of the spring member in at least two different directions to effectuate opening of the liquid-tight sealing of the valve.

(5) In some cases, the actuator member includes a head portion. Also, the first portion with which the sealing member is operatively interfaced is a first side of the head portion. Furthermore, the second portion with which the spring member is operatively interfaced is a second side of the head portion situated opposite the first side of the head portion. In some such instances, the head portion has a raised feature. Also, in being operatively interfaced with the first side of the head portion, the spring member is seated on the raised feature.

(6) In some cases, the at least two different directions are substantially perpendicular to one another. In some cases, the at least two different directions include: a first direction generally aligned with a longitudinal length of the body member; and a second direction generally aligned with a transverse width or diameter of the body member.

(7) In some cases, in terms of flow direction in the bore of the valve, the actuator member is at least one of: downstream of the spring member; and upstream of the sealing member. In some such instances, the actuator member is both: downstream of the spring member; and upstream of the sealing member.

(8) In some cases, the body member includes a piercing member having defined therein at least a portion of the bore of the valve. In some such instances, the spring member is at least partially disposed within the at least a portion of the bore defined in the piercing member. In some other such instances, the piercing member is of conical configuration, terminating in a sharp tip configured to facilitate penetration of the penetrable container by the valve.

(9) In some cases, the body member has formed therein at least one opening configured to be in flow communication with the bore to allow liquid to flow from the penetrable container into the

valve. In some cases, the body member includes a base member having defined therein at least a portion of the bore of the valve. In some such instances, the base member has at least one external projection extending radially outward therefrom. In some such instances, the at least one external projection is configured as a raised rib extending along at least a portion of a longitudinal length of the base member. In some instances, the at least one external projection terminates in a sharp end. In some instances, the at least one external projection has a sharp edge.

(10) In some cases: the body member includes a polymer; and the actuator member includes a metal. In some instances, the polymer is selected from the group consisting of: polypropylene or a polypropylene-based material; polyethylene or a polyethylene-based material; and polyoxymethylene or a polyoxymethylene-based material. In some instances, the metal is selected from the group consisting of: stainless steel or a stainless-steel based material; and copper or a copper-based material.

(11) In some cases, a system is provided, the system including the valve described herein and a cage or enclosure configured to have the valve installed thereat.

(12) The features and advantages described herein are not all-inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been selected principally for readability and instructional purposes and not to limit the scope of the inventive subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is an isometric view of a valve configured in accordance with an embodiment of the present disclosure.

(2) FIG. **2** is another isometric view of the valve of FIG. **1**.

(3) FIG. **3** is an exploded isometric view of the valve of FIG. **1**.

(4) FIG. **4** is an exploded side elevation view of the valve of FIG. **1**.

(5) FIG. **5** is a side elevation view from a first end of the valve of FIG. **1**.

(6) FIG. **6** is a side elevation view from a second end of the valve of FIG. **1**.

(7) FIG. **7** is a side elevation view of the valve of FIG. **1**.

(8) FIG. **8** is a side elevation view of the valve of FIG. **7** after rotation through 90°.

(9) FIG. **9** is a cross-sectional view of the valve of FIG. **1**.

(10) FIG. **10** is a cross-sectional view of the valve of FIG. **9** after rotation through 90°.

(11) FIG. **11** is an isometric view of a piercing member configured in accordance with an embodiment of the present disclosure.

(12) FIG. **12** is another isometric view of the piercing member of FIG. **11**.

(13) FIG. **13** is a side elevation view from a first end of the piercing member of FIG. **12**.

(14) FIG. **14** is a side elevation view from a second end of the piercing member of FIG. **12**.

(15) FIG. **15** is a side elevation view of the piercing member of FIG. **11**.

(16) FIG. **16** is a side elevation view of the piercing member of FIG. **15** after rotation through 90°.

(17) FIG. **17** is a cross-sectional view of the piercing member of FIG. **11**.

(18) FIG. **18** is a cross-sectional view of the piercing member of FIG. **17** after rotation through 90°.

(19) FIG. **19** is an isometric view of a base member configured in accordance with an embodiment of the present disclosure.

(20) FIG. **20** is another isometric view of the base member of FIG. **19**.

(21) FIG. **21** is a side elevation view from a first end of the base member of FIG. **19**.

(22) FIG. **22** is a side elevation view from a second end of the base member of FIG. **19**.

(23) FIG. **23** is a side elevation view of the base member of FIG. **19**.

(24) FIG. **24** is a cross-sectional view of the base member of FIG. **19**.
(25) FIG. **25** is an isometric cross-sectional view of the base member of FIG. **19**.
(26) FIG. **26** is an isometric view of an actuator member configured in accordance with an embodiment of the present disclosure.
(27) FIG. **27** is another isometric view of the actuator member of FIG. **26**.
(28) FIG. **28** is a side elevation view of the actuator member of FIG. **26**.
(29) FIGS. **29**A-**29**D are views of an installed valve being operatively interfaced with a penetrable container in the context of an example host platform in accordance with an embodiment of the present disclosure.
(30) FIGS. **30**A-**30**B are views of an actuated (e.g., opened) valve in accordance with an embodiment of the present disclosure.
(31) These and other features of the present embodiments will be understood better by reading the following detailed description, taken together with the figures herein described. In the drawings, each identical or nearly identical component that is illustrated in various figures may be represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing. Furthermore, as will be appreciated in light of this disclosure, the accompanying drawings are not intended to be drawn to scale or to limit the described embodiments to the specific configurations shown.

DETAILED DESCRIPTION

(32) An improved animal drinking valve is disclosed. The disclosed valve may be configured to penetrate a penetrable container containing a liquid for consumption by an animal. The disclosed valve also may include an internal actuator assembly including a spring member that biases an actuator member into liquid-tight interfacing with a sealing member situated within a bore of the valve. In this arrangement, the actuator assembly may provide the valve with two-way actuation; that is, the valve may be openable to allow liquid to flow therethrough when sufficient force is applied in directions perpendicular and/or parallel to a distal end of the actuator member. Also, in accordance with some embodiments, the disclosed valve may be of dual-material construction, having a plastic body and a metal actuator member, improving resistance to chewing by animals and other physical degradation. Furthermore, in accordance with some embodiments, the disclosed valve optionally may include one or more exterior fin-like projections that facilitate installation of the valve in a host platform. In accordance with some embodiments, the disclosed valve may be configured for use, for example, with reusable/disposable cages or other enclosures typically used for small animals, such as rodents (e.g., mice, rats, hamsters, guinea pigs, etc.). Numerous configurations and variations will be apparent in light of this disclosure.

General Overview

(33) Existing small animal drinking valves normally are constructed from only a single material, either entirely plastic or entirely metal, with the material selection typically being dictated by a balance between animal safety and valve replacement costs. At least one existing small animal drinking valve includes a plastic internal activator rod that is susceptible to chewing (or other deforming) by animal(s), which can render the valve functionless. In such cases, if the damaged valve is not replaced in a timely manner, the animal(s) may risk dehydration and death.
(34) Thus, and in accordance with some embodiments of the present disclosure, an improved animal drinking valve is disclosed. The disclosed valve may be configured to penetrate a penetrable container containing a liquid for consumption by an animal. The disclosed valve also may include an internal actuator assembly including a spring member that biases an actuator member into liquid-tight interfacing with a sealing member situated within a bore of the valve. In this arrangement, the actuator assembly may provide the valve with two-way actuation; that is, the valve may be openable to allow liquid to flow therethrough when sufficient force is applied in directions perpendicular and/or parallel to a distal end of the actuator member. Also, in accordance with some embodiments, the disclosed valve may be of dual-material construction, having a plastic

body and a metal actuator member, improving resistance to chewing by animals and other physical degradation. Furthermore, in accordance with some embodiments, the disclosed valve optionally may include one or more exterior fin-like projections that facilitate installation of the valve in a host platform. In accordance with some embodiments, the disclosed valve may be configured for use, for example, with reusable/disposable cages or other enclosures typically used for small animals, such as rodents (e.g., mice, rats, hamsters, guinea pigs, etc.).

(35) Some embodiments of the disclosed valve may realize advantages and benefits as compared to existing devices and approaches. For example, some embodiments disclosed herein may have improved durability and resistance to chewing by animals as compared to existing approaches. Some embodiments disclosed herein may be constructed solely from materials which are food-grade and safe for animals. Some embodiments disclosed herein may be substantially drip-free, thus conserving water (or other liquid) and keeping the host platform drier. Some embodiments disclosed herein may function without first requiring priming. Some embodiments disclosed herein may be single-use and disposable. Some embodiments disclosed herein may help to minimize (or otherwise reduce) cross-contamination.

(36) Structure and Operation

(37) FIGS. **1-10** illustrate several views of an animal drinking valve **100** configured in accordance with an embodiment of the present disclosure. As can be seen, valve **100** may include a piercing member **110** and a base member **120** configured to be assembled with one another in temporary or permanent manner, generally defining a body member of valve **100**. Also, as can be seen, valve **100** may include an actuator member **130**, a spring member **140**, and a sealing member **150**, generally defining an internal assembly of valve **100**. As described herein, members **130**, **140**, and **150** may be configured to be operatively interfaced with one another to effectuate repeatable opening and closing (e.g., unsealing and sealing, respectively) of valve **100**, thus controlling the ability of a liquid to flow through valve **100** in operation thereof. Each of these elements is discussed in turn below.

(38) FIGS. **11-18** illustrate several views of a piercing member **110** configured in accordance with an embodiment of the present disclosure. As can be seen, piercing member **110**, sometimes alternatively referred to as a barb or barbed tip, may include an upper body portion **112** and a lower body portion **114**. Also, as can be seen, piercing member **110** may have formed therein one or more openings **115** extending into an interior of piercing member **110** and a bore **113** extending along at least a portion of the longitudinal extent (e.g., length) of piercing member **110** from upper body portion **112** to lower body portion **114**. In some cases, upper body portion **112** optionally may include one or more concave contours or channels (e.g., which may be U-shaped, V-shaped, or C-shaped in cross-section) extending along all or some portion of the longitudinal extent (e.g., length) of its exterior surface and substantially aligning with one or more of openings **115**.

(39) In general, piercing member **110** may be configured to penetrate (e.g., pierce or puncture) a penetrable container **10** (e.g., pouch, sack, bladder, bag, etc.) containing water or other liquid(s) for consumption by an animal. Opening(s) **115** and bore **113** together may define, at least in part, an internal fluid channel or passageway of valve **100** through which liquid may flow (e.g., from upstream penetrable container **10**) when being dispensed by valve **100**.

(40) As can be seen from FIGS. **11-18**, in some embodiments, upper body portion **112** may be of generally conical shape, including (e.g., terminating in) a sharp tip **116** configured to penetrate a penetrable container **10**. In some other embodiments, however, upper body portion **112** may be of generally pyramidal shape (e.g., having three, four, or more lateral faces), again including (e.g., terminating in) a sharp tip **116**.

(41) Also, the longitudinal extent (e.g., length) and transverse extent (e.g., width or diameter) of upper body portion **112** may be customized, as desired. In some embodiments, upper body portion **112** may have a length in the range of about 0.5-2.0 in (e.g., about 0.5-1.0 in, about 1.0-1.5 in, about 1.5-2.0 in, or any other sub-range in the range of about 0.5-2.0 in). In some embodiments,

upper body portion **112** may have a maximum width or diameter in the range of about 0.25-0.5 in (e.g., about 0.25-0.3 in, about 0.3-0.35 in, about 0.35-0.4 in, about 0.4-0.45 in, about 0.45-0.5 in, or any other sub-range in the range of about 0.25-0.5 in).

(42) Also, as can be seen from FIGS. **11-18**, in some embodiments, lower body portion **114** may be of generally tubular shape, at least partially defining a bore **113** of piercing member **110**. In some embodiments, lower body portion **114** may be generally annular in cross-sectional shape, having a circular, elliptical, semi-circular, or semi-elliptical geometry. In some other embodiments, lower body portion **114** may be generally annular in cross-sectional shape, having a polygonal (e.g., triangular, rectangular, square, etc.) geometry.

(43) Also, the longitudinal extent (e.g., length) and transverse extent (e.g., width or diameter) of lower body portion **114** may be customized, as desired. In some embodiments, lower body portion **114** may have a length in the range of about 0.125-1.0 in (e.g., about 0.125-0.25 in, about 0.25-1.0 in, about 0.5-0.75 in, about 0.75-1.0 in, or any other sub-range in the range of about 0.125-1.0 in). In some embodiments, lower body portion **114** may have a width or diameter in the range of about 0.125-0.5 in (e.g., about 0.125-0.25 in, about 0.25-0.5 in, or any other sub-range in the range of about 0.125-0.5 in). In some embodiments, lower body portion **114** may have a substantially uniform width or diameter along its entire length. In some other embodiments, lower body portion **114** may have a non-uniform (e.g., tapered, flared, ribbed, notched, constricted, bulged, or otherwise varying) width or diameter along all or a portion of its entire length.

(44) Regarding bore **113** of piercing member **110**, the general shape thereof may be customized, as desired. In some embodiments, bore **113** may be generally circular, elliptical, semi-circular, or semi-elliptical in cross-sectional shape. In some other embodiments, bore **113** may be generally polygonal (e.g., triangular, rectangular, square, etc.) in cross-sectional shape.

(45) Also, the longitudinal extent (e.g., length) and transverse extent (e.g., width or diameter) of bore **113** may be customized, as desired. In some embodiments, bore **113** may have a length in the range of about 0.0625-0.5 in (e.g., about 0.0625-0.1 in, about 0.1-0.25 in, about 0.25-0.5 in, or any other sub-range in the range of about 0.0625-0.5 in). In some embodiments, bore **113** may have a width or diameter in the range of about 0.125-0.5 in (e.g., about 0.125-0.25 in, about 0.25-0.5 in, or any other sub-range in the range of about 0.125-0.5 in). In some embodiments, bore **113** may have a substantially uniform width or diameter along its entire length. In some other embodiments, bore **113** may have a non-uniform (e.g., tapered, flared, ribbed, notched, constricted, bulged, or otherwise varying) width or diameter along all or a portion of its entire length. As will be appreciated in light of this disclosure, it generally may be desirable to ensure that bore **113** is sufficiently sized and of appropriate geometry to permit spring member **140** to compress and decompress therein to a degree sufficient to permit fluid flow in operation of valve **100**.

(46) Furthermore, as can be seen from FIGS. **17-18**, lower body portion **114** may have a shoulder **117** formed in bore **113**. Shoulder **117** may be configured to have spring member **140** (discussed below) situated against it (e.g., directly or indirectly via one or more intervening layers), thereby facilitating retention of spring member **140** within valve **100**. To that end, the dimensions and contour(s) of shoulder **117** may be customized, as desired.

(47) Regarding opening(s) **115**, the general shape and size thereof may be customized, as desired. In some embodiments, a given opening **115** may be generally circular, elliptical, semi-circular, or semi-elliptical in cross-sectional shape. In some other embodiments, a given opening **115** may be generally polygonal (e.g., triangular, rectangular, square, etc.) in cross-sectional shape. Also, the longitudinal extent (e.g., length) and transverse extent (e.g., width or diameter) of a given opening **115** may be customized, as desired, and may depend, at least in part, on the local material thickness of upper body portion **112** and/or lower body portion **114**.

(48) In some embodiments, piercing member **110** may include only a single opening **115**, whereas in other embodiments, a plurality of openings **115** (e.g., two, three, four, or more) may be provided. In cases where a plurality of openings **115** is provided, the arrangement/spacing thereof may be

customized, as desired. For instance, any two openings **115** may be spaced apart from one another, for example, at an offset of 180°, 150°, 135°, 120°, 90°, 60°, 45°, 30°, or any other desired amount around the periphery of piercing member **110**. As can be seen from FIGS. **11-14**, in an example embodiment, piercing member **110** may include two similarly configured openings **115** arranged substantially opposite one another (e.g., at an offset of about 180°) about the periphery of piercing member **110** in the region where upper body portion **112** and lower body portion **114** meet.

(49) Piercing member **110** may be constructed, in part or in whole, from any of a wide range of suitable materials, including polymers and composites. For instance, in some cases, piercing member **110** may be formed, at least in part, from polypropylene (PP) or a polypropylene-based material. In some cases, piercing member **110** may be formed, at least in part, from polyethylene (PE) or a polyethylene-based material. In some cases, piercing member **110** may be formed, at least in part, from polyoxymethylene (POM) or a polyoxymethylene-based material. Moreover, piercing member **110** may be of monolithic construction (i.e., single-piece construction) or polylithic construction (i.e., multi-piece construction), as desired. Other suitable constructions for piercing member **110** will depend on a given target application or end-use and will be apparent in light of this disclosure.

(50) FIGS. **19-25** illustrate several views of a base member **120** configured in accordance with an embodiment of the present disclosure. As can be seen, base member **120**, sometimes alternatively referred to as a hub, may include a lower body portion **122** and an upper body portion **124**. Also, as can be seen, an exterior of base member **120** may include a flange portion **126** extending radially, for example, from the region where lower body portion **122** and upper body portion **124** meet. Furthermore, as can be seen, base member **120** may have formed therein a bore **123** extending along at least a portion of its longitudinal extent (e.g., length) from upper body portion **124** to lower body portion **122**.

(51) In general, base member **120** may be configured to be situated adjacent (e.g., in direct or indirect physical contact with or otherwise disposed proximate to) the exterior of a penetrable container **10** with which valve **100** is interfaced. In at least some cases, however, at least a portion of base member **120** (e.g., upper body portion **124**) may be configured to be at least partially inserted into penetrable container **10** in a manner providing a liquid-tight seal at the interface thereof. As will be appreciated in light of this disclosure, such liquid-tight sealing may prevent (or otherwise reduce) leakage at the installation site for valve **100**. By virtue of the physical presence of flange portion **126** (discussed below), lower body portion **122** may be prevented from being inserted into penetrable container **10**. In some cases, flange portion **26** may be made to abut the exterior of penetrable container **10**, though such abutting is not required for proper operation of valve **100**, as a gap between the exterior of penetrable container **10** and flange portion **126** may be permitted in some instances. Furthermore, bore **123** may define, at least in part, an internal fluid channel or passageway of valve **100** through which liquid may flow (e.g., from upstream piercing member **110**) when being dispensed by valve **100** and, to that end, may be configured to be in flow communication with bore **113** (of piercing member **110**).

(52) As can be seen from FIGS. **19-25**, in some embodiments, upper body portion **124** and/or lower body portion **122** may be of generally tubular shape, at least partially defining a bore **123** of base member **120**. In some embodiments, upper body portion **124** and/or lower body portion **122** may be generally annular in cross-sectional shape, having a circular, elliptical, semi-circular, or semi-elliptical geometry. In some other embodiments, upper body portion **124** and/or lower body portion **122** may be generally annular in cross-sectional shape, having a generally polygonal (e.g., triangular, rectangular, square, etc.) geometry.

(53) Also, the longitudinal extent (e.g., length) and transverse extent (e.g., width or diameter) of upper body portion **124** and/or lower body portion **122** may be customized, as desired. In some embodiments, upper body portion **124** may have a length in the range of about 0.25-0.5 in (e.g., about 0.25-0.3 in, about 0.3-0.35 in, about 0.35-0.4 in, about 0.4-0.45 in, about 0.45-0.5 in, or any

other sub-range in the range of about 0.25-0.5 in). In some embodiments, upper body portion **124** may have a width or diameter in the range of about 0.25-0.5 in (e.g., about 0.25-0.3 in, about 0.3-0.35 in, about 0.35-0.4 in, about 0.4-0.45 in, about 0.45-0.5 in, or any other sub-range in the range of about 0.25-0.5 in). In some embodiments, lower body portion **122** may have a length in the range of about 0.25-1.0 in (e.g., about 0.25-0.5 in, about 0.5-0.75 in, about 0.75-1.0 in, or any other sub-range in the range of about 0.25-1.0 in). In some embodiments, lower body portion **122** may have a width or diameter in the range of about 0.25-0.5 in (e.g., about 0.25-0.3 in, about 0.3-0.35 in, about 0.35-0.4 in, about 0.4-0.45 in, about 0.45-0.5 in, or any other sub-range in the range of about 0.25-0.5 in). In some embodiments, upper body portion **124** and/or lower body portion **122** may have a substantially uniform width or diameter along its entire length. In some other embodiments, upper body portion **124** and/or lower body portion **122** may have a non-uniform (e.g., tapered, flared, ribbed, notched, constricted, bulged, or otherwise varying) width or diameter along all or a portion of its entire length. In some embodiments, upper body portion **124** and lower body portion **122** may be of substantially similar (e.g., the same) shape and/or size, whereas in other embodiments, they may differ from one another in one or more aspects.

(54) Regarding bore **123** of base member **120**, the general shape thereof may be customized, as desired. In some embodiments, bore **123** may be generally circular, elliptical, semi-circular, or semi-elliptical in cross-sectional shape. In some other embodiments, bore **123** may be generally polygonal (e.g., triangular, rectangular, square, etc.) in cross-sectional shape.

(55) Also, the longitudinal extent (e.g., length) and transverse extent (e.g., width or diameter) of bore **123** may be customized, as desired. In some embodiments, the segment of bore **123** within upper body portion **124** may have a length in the range of about 0.0625-0.5 in (e.g., about 0.0625-0.1 in, about 0.1-0.25 in, about 0.25-0.5 in, or any other sub-range in the range of about 0.0625-0.5 in). In some embodiments, the segment of bore **123** within upper body portion **124** may have a width or diameter in the range of about 0.25-0.5 in (e.g., about 0.25-0.3 in, about 0.3-0.35 in, about 0.35-0.4 in, about 0.4-0.45 in, about 0.45-0.5 in, or any other sub-range in the range of about 0.25-0.5 in). In some embodiments, the segment of bore **123** within lower body portion **122** may have a width or diameter in the range of 1-5 mm (e.g., 1-2.5 mm, 2.5-3.5 mm, 3.5-5 mm, or any other sub-range in the range of 1-5 mm). In some embodiments, bore **123** may have a substantially uniform width or diameter along its entire length. In some other embodiments, bore **123** may have a non-uniform (e.g., tapered, flared, ribbed, notched, constricted, bulged, or otherwise varying) width or diameter along all or a portion of its entire length. As will be appreciated in light of this disclosure, it generally may be desirable to ensure that bore **123** is sufficiently sized and of appropriate geometry to permit actuator member **130** to move longitudinally and/or laterally therein to a degree sufficient to permit proper sealing/unsealing of the fluid channel of valve **100** and, thus, fluid flow in operation of valve **100**. Also, it generally may be desirable to ensure that bore **123** is not so large as to permit lodging of debris therein, as such foreign matter could impede or negate the function of actuator member **130** and, thus, render valve **100** inoperable.

(56) Furthermore, as can be seen from FIGS. **24-25**, base member **120** may have a shoulder **127** formed in bore **123** in the region where lower body portion **122**, upper body portion **124**, and flange portion **126** meet. Shoulder **127** may be configured to have sealing member **150** (discussed below) situated against it (e.g., directly or indirectly via one or more intervening layers), thereby facilitating retention of sealing member **150** within valve **100**. To that end, the dimensions and contour(s) of shoulder **127** may be customized, as desired.

(57) In accordance with some embodiments, base member **120** optionally may include one or more exterior projections **122*a*** configured to facilitate provision of a secure fit between valve **100** and a given host platform **20** (e.g., a reusable/disposable cage or other enclosure). For instance, projection(s) **122*a*** may be configured to puncture, pierce, cut, or slice through the local material at a given installation site of host platform **20**. If host platform **20** includes, for example, a pre-formed perforation, cross-cut, hole, or other localized thinning or weakening of material, projection(s)

122*a* may facilitate insertion and retention of valve **100** at such installation site. Additionally (or alternatively), if a rubber grommet (or the like) is included at the installation site, projection(s) **122***a* may be interfaced therewith, providing a reliable friction fit for valve **100** at such installation site.
(58) As can be seen from FIGS. **19-23**, in some embodiments, a given projection **122***a* may be a raised fin or rib extending along all or at least a portion of the longitudinal extent (e.g., length) of lower body portion **122**. In some embodiments, a given projection **122***a* may be continuous in form along its entire length, whereas in other embodiments, a given projection **122***a* may be segmented or otherwise discontinuous in form along all or a portion of its entire length. In some embodiments, a given projection **122***a* may extend substantially radially outward from the exterior surface of lower body portion **122**. In some instances, a given projection **122***a* may extend all the way out to the perimetral edge of flange portion **126**, whereas in other instances, a given projection **122***a* may terminate radially inward of that perimetral edge.
(59) In some embodiments, a given projection **122***a* may terminate in a chamfered, angled, or beveled edge, the profile of which may be customized, as desired. In some cases, the end of a given projection **122***a* may terminate in a sharp or pointed edge sufficient to effectuate cutting or piercing. In some cases, a given projection **122***a* may be generally knife-like or razor-like in cutting profile.
(60) Regarding projection(s) **122***a*, the general shape and size thereof may be customized, as desired. As can be seen from FIGS. **19-23**, in some embodiments, a given projection **122***a* may be generally prismatic in shape, having a generally polygonal (e.g., triangular, rectangular, square, etc.) cross-sectional geometry. In some other embodiments, a given projection **122***a* may be generally cylindrical in shape, having a generally rounded (e.g., circular, elliptical, semi-circular, semi-elliptical, etc.) cross-sectional geometry. Also, the longitudinal extent (e.g., length), transverse extent (e.g., width or diameter), and radial extent of a given projection **122***a* may be customized, as desired, and may depend, at least in part, on the dimension(s) of lower body portion **122** and/or flange portion **126** (discussed below).
(61) In some embodiments, base member **120** may include only a single projection **122***a*, whereas in other embodiments, a plurality of projections **122***a* (e.g., two, three, four, or more) may be provided. In cases where a plurality of projections **122***a* is provided, the arrangement/spacing thereof may be customized, as desired. For instance, any two projections **122***a* may be spaced apart from one another, for example, at an offset of 180°, 150°, 135°, 120°, 90°, 60°, 45°, 30°, or any other desired amount around the periphery of lower body portion **122**. As can be seen from FIGS. **19-22**, in an example embodiment, base member **120** may include three similarly configured projections **122***a* arranged substantially radially equidistant from one another (e.g., at an offset of about) 120° about the periphery of lower body portion **122**.
(62) Regarding flange portion **126**, the general shape and size thereof may be customized, as desired. As can be seen from FIGS. **19-25**, in some embodiments, flange portion **126** may be generally disc-like, plate-like, or ring-like, having a generally circular, elliptical, semi-circular, or semi-elliptical geometry. In some other embodiments, flange portion **126** may be generally disc-like, plate-like, or ring-like, having a generally polygonal (e.g., triangular, rectangular, square, etc.) geometry. In some embodiments, flange portion **126** may be of substantially planar form, whereas in other embodiments, flange portion **126** may have one or more regions of non-planarity (e.g., faceted, undulating, textured, or stepped surface contour). Also, the longitudinal extent (e.g., thickness) and transverse extent (e.g., width or diameter) of flange portion **126** may be customized, as desired.
(63) Base member **120** may be constructed, in part or in whole, from any of a wide range of suitable materials. For instance, in some cases, base member **120** may be formed, at least in part, from polypropylene (PP) or a polypropylene-based material. In some cases, base member **120** may be formed, at least in part, from polyethylene (PE) or a polyethylene-based material. In some cases,

base member **120** may be formed, at least in part, from polyoxymethylene (POM) or a polyoxymethylene-based material. In some cases, base member **120** may be formed, at least in part, from stainless steel or a stainless steel-based material. Moreover, base member **120** may be of monolithic construction (i.e., single-piece construction) or polylithic construction (i.e., multi-piece construction), as desired. Other suitable constructions for base member **120** will depend on a given target application or end-use and will be apparent in light of this disclosure.
(64) In construction of valve **100**, piercing member **110** and base member **120** may be engaged or otherwise interfaced with one another in a temporary or permanent manner, as desired. To that end, lower body portion **114** (of piercing member **110**) and upper body portion **124** (of base member **120**) may include corresponding engagement features **118**, **128**. In some embodiments, engagement feature **118** (of piercing member **110**) may be a protruding rib, and engagement feature **128** (of base member **120**) may be a recessed channel which is to receive and retain that protruding rib, providing for snap-fit engagement therebetween. In other embodiments, piercing member **110** and base member **120** may be configured for threaded or friction-fit engagement with one another. In still other embodiments, piercing member **110** and base member **120** may be configured for mechanical fastener-based engagement or adhesive engagement with one another. Other suitable configurations for assembling members **110**, **120** will depend on a given target application or end-use and will be apparent in light of this disclosure.
(65) As noted above, valve **100** also may include an internal assembly of components including an actuator member **130**, a spring member **140**, and a sealing member **150**, which may be configured to work in concert with one another within bores **113**, **123** to effectuate sealing/unsealing of the fluid channel in operation of valve **100**. Each of these elements is discussed in turn below.
(66) FIGS. **26-28** illustrate several views of an actuator member **130** configured in accordance with an embodiment of the present disclosure. As can be seen, actuator member **130**, sometimes alternatively referred to as a stem, may include a body portion **132** having a head portion **134** at a first end **131** thereof.
(67) In general, actuator member **130** may be configured to be situated within bore **123** (of base member **120**). In such arrangement, head portion **134** may reside within the segment of bore **123** within upper body portion **124**, while body portion **132** may reside in part within that same segment and in part within the segment of bore **123** within lower body portion **122**. As can be seen further, a second end **133** of body portion **132** may extend beyond the longitudinal extent (e.g., length) of lower body portion **122**, thus remaining easily accessible to an animal or other actor intending to operate valve **100**.
(68) As can be seen from FIGS. **26-28**, in some embodiments, body portion **132** may be generally configured as a rod, shaft, or other rigid or semi-rigid, elongated body. In some embodiments, body portion **132** may be generally cylindrical in shape, having a circular, elliptical, semi-circular, or semi-elliptical cross-sectional geometry. In some other embodiments, body portion **132** may be generally prismatic in shape, having a polygonal (e.g., triangular, rectangular, square, etc.) cross-sectional geometry.
(69) Also, the longitudinal extent (e.g., length) and transverse extent (e.g., width or diameter) of body portion **132** may be customized, as desired. In some embodiments, body portion **132** may have a length in the range of about 0.125-2.0 in (e.g., about 0.125-0.25 in, about 0.25-0.5 in, about 0.5-0.75 in, about 0.75-1.0 in, about 1.0-1.25 in, about 1.25-1.5 in, about 1.5-1.75 in, about 1.75-2.0 in or any other sub-range in the range of about 0.125-2.0 in). In some embodiments, body portion **132** may have a width or diameter in the range of about 0.0625-0.125 in (e.g., about 0.0625-0.08 in, about 0.08-0.1 in, about 0.1-0.125 in, or any other sub-range in the range of about 0.0625-0.125 in). In some embodiments, body portion **132** may have a substantially uniform width or diameter along its entire length. In some other embodiments, body portion **132** may have a non-uniform (e.g., tapered, flared, ribbed, notched, constricted, bulged, or otherwise varying) width or diameter along all or a portion of its entire length. As will be appreciated in light of this disclosure,

it generally may be desirable to ensure that body portion **132** is sufficiently sized to permit actuator member **130** to move longitudinally and/or laterally within bore **123** to a degree sufficient to permit fluid flow in operation of valve **100**.

(70) As can be seen from FIGS. **26-28**, in some embodiments, head portion **134** may be generally configured as a disc, plate, or puck-like body. In some embodiments, head portion **134** may be generally circular, elliptical, semi-circular, or semi-elliptical in shape. In some other embodiments, head portion **134** may be generally polygonal (e.g., triangular, rectangular, square, etc.) in shape. As can be seen further, in some embodiments, head portion **134** may include a raised feature **136** projecting outward from a first side **135** thereof (opposite a second side **137** from which body portion **132** extends). In some such instances, raised feature **136** may include one or more sloped or stepped shoulders, the angle/grade of which may be customized, as desired.

(71) Also, the longitudinal extent (e.g., thickness) and transverse extent (e.g., width or diameter) of head portion **134** may be customized, as desired. In some embodiments, head portion **134** may have a width or diameter commensurate with a width or diameter of upper body portion **124**. In some embodiments, head portion **134** may have a substantially uniform thickness. In some other embodiments, head portion **134** may have a non-uniform (e.g., tapered, flared, ribbed, notched, constricted, bulged, or otherwise varying) thickness. As will be appreciated in light of this disclosure, it generally may be desirable to ensure that head portion **134** is sufficiently sized to permit actuator member **130** to move longitudinally and/or laterally within bore **123** to a degree sufficient to permit fluid flow in operation of valve **100**.

(72) Actuator member **130** may be constructed, in part or in whole, from any of a wide range of suitable materials, including metals or alloys. For instance, in some cases, actuator member **130** may be formed, at least in part, from a stainless steel or a stainless steel-based material, such as 302 stainless steel or 304 stainless steel, among other suitable types. In some cases, actuator member **130** may be formed, at least in part, from copper (Cu) or a copper-based material, such as brass. Moreover, actuator member **130** may be of monolithic construction (i.e., single-piece construction) or polylithic construction (i.e., multi-piece construction), as desired. Other suitable constructions for actuator member **130** will depend on a given target application or end-use and will be apparent in light of this disclosure.

(73) As previously noted, the internal assembly of valve **100** also may include a spring member **140**. As can be seen from FIGS. **3-4** and **9-10**, in some embodiments, spring member **140** may be, for example, a compression coil spring. It should be noted, however, that the present disclosure is not intended to be limited only to a coil spring, as in a more general sense, and in accordance with some other embodiments, any suitable spring or other member with sufficient resilience may be utilized, as will be apparent in light of this disclosure.

(74) In general, spring member **140** may be configured to be situated in part within bore **113** (of piercing member **110**) and in part within the segment of bore **123** within upper body portion **124** (of base member **120**). In such arrangement, a first end of spring member **140** may be made to abut (e.g., directly or indirectly via one or more intervening layers) shoulder **117** of piercing member **110**, while a second end of spring member **140** may be made to abut (e.g., directly or indirectly via one or more intervening layers) head portion **134** of actuator member **130**, for example, at raised feature **136** thereof. In this manner, raised feature **136** may serve, at least in part, to seat spring member **140** on head portion **134**. Furthermore, in such arrangement, spring member **140** may be compressed against shoulder **117** when sufficient force is applied to actuator member **130** to cause deflection thereof and allowed to decompress (or remain uncompressed) when insufficient force (e.g., no force) is applied. In this manner, spring member **140** may be configured to bias actuator member **130** into liquid-tight sealing contact (e.g., directly or indirectly via one or more intervening layers) with sealing member **150**.

(75) Also, the dimensions (e.g., wire diameter, coil diameter, and overall length) and stiffness of spring member **140** may be customized, as desired. In some embodiments, spring member **140** may

have a coil diameter in the range of about 0.25-0.75 mm (e.g., about 0.25-0.5 mm, about 0.5-0.75 mm, or any other sub-range in the range of about 0.25-0.75 mm). In a specific example case, spring member **140** may have a wire diameter of 0.3 mm, a coil diameter of 0.5 mm, and an overall length of 12 mm. In a specific example case, spring member **140** may be a 2-gram spring. As will be appreciated in light of this disclosure, it generally may be desirable to ensure that spring member **140** is sufficiently constructed and arranged to permit actuator member **130** to move longitudinally and/or laterally within bore **123** to a degree sufficient to permit fluid flow in operation of valve **100**.

(76) Spring member **140** may be constructed, in part or in whole, from any of a wide range of suitable materials. For instance, in some cases, spring member **140** may be formed, at least in part, from stainless steel or a stainless steel-based material, such as 302 stainless steel or 304 stainless steel, among other suitable types. Moreover, spring member **140** may be of monolithic construction (i.e., single-piece construction) or polylithic construction (i.e., multi-piece construction), as desired. Other suitable constructions for spring member **140** will depend on a given target application or end-use and will be apparent in light of this disclosure.

(77) As previously noted, the internal assembly of valve **100** also may include a sealing member **150**. As can be seen from FIGS. **3-4** and **9-10**, in some embodiments, sealing member **150** may be, for example, an O-ring or other suitable gasket. As can be seen further, sealing member **150** may have an opening (e.g., a central aperture or hole) through which body portion **132** of actuator member **130** may be passed.

(78) In general, sealing member **150** may be configured to be situated within the segment of bore **123** within upper body portion **124** (of base member **120**). In such arrangement, a first side of sealing member **150** may be made to abut (e.g., directly or indirectly via one or more intervening layers) shoulder **127** of base member **120**, while a second side of sealing member **150** may be made to abut (e.g., directly or indirectly via one or more intervening layers) head portion **134** of actuator member **130**, for example, at a second side **137** thereof. In this manner, sealing member **150** may be compressed against shoulder **127** when insufficient force (e.g., no force) is applied to actuator member **130** to cause deflection thereof and allowed to decompress (or remain uncompressed) when sufficient force is applied. Consequently, sealing member **150** may be biased into contact with base member **120** (via spring member **140** through intervening actuator member **130**) within bore **123** thereof, thereby effectuating a liquid-tight sealing of the fluid channel of valve **100**.

(79) Also, the shape and hardness rating of sealing member **150** may be customized, as desired. In some embodiments, sealing member **150** may be of generally toroidal or donut shape (e.g., an O-ring shape), having body portion **132** of actuator member **130** inserted therethrough. In a specific example case, sealing member **150** may be of a 55 A hardness rating, though other suitable hardness ratings will be apparent in light of this disclosure and will depend upon a given target application or end-use.

(80) Sealing member **150** may be constructed, in part or in whole, from any of a wide range of suitable materials. For instance, sealing member **150** may be formed, at least in part, from an elastomeric material, such as a rubber (e.g., natural or synthetic) or a thermoplastic, among others. Moreover, sealing member **150** may be of monolithic construction (i.e., single-piece construction) or polylithic construction (i.e., multi-piece construction), as desired. Other suitable constructions for sealing member **150** will depend on a given target application or end-use and will be apparent in light of this disclosure.

(81) FIGS. **29**A-**29**D illustrate several views of an installed valve **100** being operatively interfaced with a penetrable container **10** in the context of an example host platform **20** in accordance with an embodiment of the present disclosure. As can be seen from FIG. **29**A, for example, valve **100** first may be installed at an installation site of host platform **20**. To such ends, valve **100** may be interfaced with host platform **20**, such as by pressing lower body portion **122** (optionally with one or more exterior projections **122***a*) into the installation site. In so doing, flange portion **126** may

come to rest on (or near) the host wall or other surface of host platform **20**. As previously noted, host platform **20** may be, for example, a reusable/disposable cage or other enclosure, among other options.

(82) As can be seen from FIGS. **29**B-**29**C, for example, penetrable container **10** then may be placed into contact with valve **100**, which in turn may be forced or otherwise permitted to penetrate penetrable container **20** to access a liquid contained therein. As can be seen from FIG. **29**D, for example, valve **100** may be at least partially inserted into penetrable container **10** (e.g., up to flange portion **126**). In this position, lower body portion **122** of base member **120** may remain accessible outside of penetrable container **10**, and, thus, actuator member **130** may remain accessible for actuation by an animal or other actor in normal operation of valve **100**.

(83) Regarding operation of valve **100** to effectuate a liquid flow therethrough, consider FIGS. **30**A-**30**B, which illustrate several views of an actuated (e.g., opened) valve in accordance with an embodiment of the present disclosure. As can be seen from FIG. **30**A, for example, a force may be applied to a second end **133** of actuator member **130** to cause deflection thereof (from its normal resting position against the restorative force of spring member **140**) in a direction aligned, to a given degree, with a transverse extent (e.g., width or diameter) of base member **120** (or the overall body member of valve **100** more generally). That is, application of a sufficient force in the X-direction seen in FIG. **30**A may cause actuator member **130** to at least partially separate from sealing member **150**, thereby opening valve **100** and allowing a liquid to flow therethrough. As can be seen from FIG. **30**B, for example, a force alternatively (or additionally) may be applied to a second end **133** of actuator member **130** to cause deflection thereof (from its normal resting position against the restorative force of spring member **140**) in a direction aligned, to a given degree, with a longitudinal extent (e.g., length) of base member **120** (or the overall body member of valve **100** more generally). That is, application of a sufficient force in the Y-direction seen in FIG. **30**B may cause actuator member **130** to at least partially separate from sealing member **150**, thereby opening valve **100** and allowing a liquid to flow therethrough. In this manner, and in accordance with some embodiments, valve **100** may be configured for two-way actuation: side-to-side movement of actuator member **130** and up-and-down movement of actuator member **130**. Otherwise stated, actuator member **130** may be deflected horizontally and/or vertically in relation to its typical resting position within bore **123** to adjust the sealing state of valve **100**. Moreover, in accordance with some embodiments, actuation also may occur in instances when a force having components in both the X-direction and Y-direction (as seen in FIGS. **30**A-**30**B) is applied to actuator member **130**.

(84) The foregoing description of example embodiments has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed. Many modifications and variations are possible in light of this disclosure. It is intended that the scope of the present disclosure be limited not by this detailed description. Future-filed applications claiming priority to this application may claim the disclosed subject matter in a different manner and generally may include any set of one or more limitations as variously disclosed or otherwise demonstrated herein.

Claims

1. An animal drinking valve configured to penetrate a penetrable container to allow a liquid to flow from the penetrable container through the valve, the valve comprising: a body member; and an assembly disposed within a bore of the body member, the assembly comprising: a sealing member; an actuator member extending through the sealing member; and a spring member operatively interfaced with a first portion of the actuator member so as to bias a second portion of the actuator member toward the sealing member to effectuate a liquid-tight sealing of the valve; wherein the actuator member is movable within the bore against the bias of the spring member in at least two

different directions to effectuate opening of the liquid-tight sealing of the valve.
2. The valve of claim 1, wherein: the actuator member comprises a head portion; the first portion with which the sealing member is operatively interfaced is a first side of the head portion; and the second portion with which the spring member is operatively interfaced is a second side of the head portion situated opposite the first side of the head portion.
3. The valve of claim 2, wherein: the head portion has a raised feature; and in being operatively interfaced with the first side of the head portion, the spring member is seated on the raised feature.
4. The valve of claim 1, wherein the at least two different directions are substantially perpendicular to one another.
5. The valve of claim 1, wherein the at least two different directions include: a first direction generally aligned with a longitudinal length of the body member; and a second direction generally aligned with a transverse width or diameter of the body member.
6. The valve of claim 1, wherein in terms of flow direction in the bore of the valve, the actuator member is at least one of: downstream of the spring member; and upstream of the sealing member.
7. The valve of claim 6, wherein the actuator member is both: downstream of the spring member; and upstream of the sealing member.
8. The valve of claim 1, wherein the body member comprises a piercing member having defined therein at least a portion of the bore of the valve.
9. The valve of claim 8, wherein the spring member is at least partially disposed within the at least a portion of the bore defined in the piercing member.
10. The valve of claim 8, wherein the piercing member is of conical configuration, terminating in a sharp tip configured to facilitate penetration of the penetrable container by the valve.
11. The valve of claim 1, wherein the body member has formed therein at least one opening configured to be in flow communication with the bore to allow liquid to flow from the penetrable container into the valve.
12. The valve of claim 1, wherein the body member comprises a base member having defined therein at least a portion of the bore of the valve.
13. The valve of claim 12, wherein the base member has at least one external projection extending radially outward therefrom.
14. The valve of claim 13, wherein the at least one external projection is configured as a raised rib extending along at least a portion of a longitudinal length of the base member.
15. The valve of claim 13, wherein the at least one external projection terminates in a sharp end.
16. The valve of claim 13, wherein the at least one external projection has a sharp edge.
17. The valve of claim 1, wherein: the body member comprises a polymer; and the actuator member comprises a metal.
18. The valve of claim 17, wherein the polymer is selected from the group consisting of: polypropylene or a polypropylene-based material; polyethylene or a polyethylene-based material; and polyoxymethylene or a polyoxymethylene-based material.
19. The valve of claim 17, wherein the metal is selected from the group consisting of: stainless steel or a stainless-steel based material; and copper or a copper-based material.
20. A system comprising: the valve of claim 1; and a cage or enclosure configured to have the valve installed thereat.